

PROJECT PROPOSAL & RESEARCH PLAN

REVIEW-1

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs

An offline, edge-optimized medical AI assistant for explainable chest X-ray interpretation

Project Group	Group 4 Cluster 3 Section 7
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Domain	Medical AI / Computer Vision / Vision-Language Models
Deployment Goal	Offline, edge-optimized assistant for rural healthcare settings

1. Project Title

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs

An offline, edge-optimized medical AI solution that combines automated chest X-ray interpretation, visual explanation through heatmaps and attention overlays, plain-language diagnostic descriptions, and conservative priority triage guidance for rural health workers.

2. Problem Statement

Primary health centres in rural regions may face limited access to specialist radiologists. This can delay interpretation of chest radiographs and increase the need for patients to travel to urban hospitals. Existing AI systems can provide disease classifications, but a prediction without a clear visual or textual explanation can be difficult for non-specialist healthcare workers to trust and interpret.

- Acute radiologist shortage: rural health centres may not have immediate specialist interpretation.
- Diagnostic delays and travel burden: patients may need to travel to urban hospitals for specialist review.
- Black-box AI mistrust: class probabilities alone do not show why a model reached its conclusion.
- Connectivity constraints: cloud-only AI is unsuitable for locations with unreliable or absent internet.

The proposed project addresses these issues by building an explainable multimodal assistant that accepts a chest X-ray, identifies likely thoracic abnormalities, highlights relevant image regions, generates a simple diagnostic description, and assigns a conservative priority/urgency flag. The system is intended as decision support, not as a replacement for a clinician.

3. Objectives

Objective	Description
Multimodal Diagnosis	Develop a vision-language assistant that produces disease predictions/confidence scores together with plain-language diagnostic descriptions.
Visual Explanation	Generate Grad-CAM heatmaps and attention overlays that identify image regions contributing to the prediction.
Automated Triage & Priority Guidance	Convert validated model outputs into a conservative urgency flag such as routine, priority, or urgent specialist review.
Offline Edge Deployment	Use model optimization and INT8 quantization to target execution on a mid-range Android device without continuous internet connectivity.
Evaluation	Measure classification quality, explanation/localization quality, text-generation quality, and mobile runtime performance.

4. Proposed Model / Framework

The proposed framework is a modular dual-stream vision-language architecture. It combines a ViT-based vision encoder for spatial feature extraction, CLIP-style/medical contrastive alignment for image-text representation, and a BLIP-2-inspired generation component for concise diagnostic text. Grad-CAM and attention visualization provide explainability, while INT8 quantization and ONNX/mobile runtime conversion support offline deployment.

1. Input Layer — chest radiograph (PNG/JPEG/DICOM converted to the required model format).
2. Pre-processing Layer — resizing, normalization, quality checks and medically appropriate training augmentation.
3. Vision Encoder — ViT-based image feature extraction for spatial representations.
4. Vision-Text Alignment — CLIP-style/medical contrastive alignment into a shared image-text space.

5. Disease Prediction Head — multi-label thoracic disease classification with calibrated confidence scores.
6. Language Generation Head — BLIP-2-inspired image-to-text generation for a concise plain-language description.
7. Explainability Layer — class-specific Grad-CAM heatmaps and cross-attention visualization where supported.
8. Triage Layer — rule-based conservative priority mapping from validated outputs.
9. Optimization Layer — pruning/precision reduction as appropriate, followed by INT8 quantization and ONNX/mobile conversion.
10. Application Layer — Android interface showing X-ray, predictions, confidence, explanation heatmap, generated description and triage flag.

4.1 How the Framework Addresses Existing Limitations

Existing Limitation	Proposed Response
Classification-only systems	Add language generation and multimodal reasoning rather than returning only labels.
Poor transparency	Add Grad-CAM and attention overlays tied to the predicted abnormality.
Paired-data dependency	Use both paired image-report datasets and compatible image-only datasets at appropriate training stages.
Heavy cloud inference	Optimize and quantize the final model for offline edge execution.
Lack of actionable prioritization	Add a separated triage-support module based on validated thresholds and safety rules.

5. Literature Survey

The literature survey below is based on the supplied project report and includes the ten works requested for the project proposal. The survey covers foundational image architectures, chest-X-ray classification, explainability, medical vision-language learning, and large-scale chest radiograph datasets.

N o.	Research Paper Title	Model / Framework	Outcome / Key Contribution	Limitations / Research Gap	Datasets Used	Paper Link
1	Deep Residual Learning for Image Recognition	ResNet / residual blocks	Introduced residual learning with shortcut connections, enabling deeper networks and strong image-recognition performance.	General computer-vision model; not designed for multimodal radiology, report generation, explainability or triage.	ImageNet, CIFAR-10 and other recognition benchmarks	https://arxiv.org/abs/1512.03385
2	Densely Connected Convolutional Networks	DenseNet	Dense connections improve feature propagation and feature reuse while reducing parameter requirements relative to comparable deep networks.	Primarily an image-recognition architecture; does not inherently provide language reasoning, report generation or clinical triage.	CIFAR-10, CIFAR-100, SVHN, ImageNet	https://arxiv.org/abs/1608.06993
3	CheXNet: Radiologist-Level	121-layer DenseNet / CheXNet	Detected pneumonia from	Primarily classification/localization; lacks	ChestX-ray14; separately	https://arxiv.org/abs/1711.05225

	Pneumonia Detection on Chest X-Rays with Deep Learning		chest radiograph s and extended the model to multiple ChestX-ray14 disease labels.	complete vision-language diagnostic conversation, natural-language explanation and offline triage workflow.	annotated radiologist test set of 420 frontal chest X-rays	
4	Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization	Grad-CAM	Produces class-discriminative heatmaps identifying image regions important for a prediction.	Provides visual attribution but does not itself generate clinical textual explanation, severity assessment or triage recommendation.	Multiple image-classification, captioning and VQA examples/datasets	https://arxiv.org/abs/1610.02391
5	BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models	BLIP-2 with Q-Former	Bridges frozen image encoders and frozen large language models using a lightweight Querying Transformer.	Large vision-language components can still be computationally demanding for small offline devices; not specifically designed for chest-radiograph safety or triage.	VQAv2 and multiple vision-language benchmark/image-text generation tasks	https://arxiv.org/abs/2301.12597
6	MedKLIP: Medical Knowledge Enhanced Language-Image Pre-Training in Radiology	MedKLIP	Uses medical knowledge, extracted medical entities and spatial alignment to strengthen radiology image-language pre-training; evaluated on disease classification and grounding.	More complex and computationally demanding than lightweight edge-oriented architectures; does not directly solve offline mobile deployment and triage.	ChestX-ray14, RSNA Pneumonia, SIIM-ACR Pneumothorax, COVIDx CXR-2, COVID Rural, EdemaSeverity	https://arxiv.org/abs/2301.02228
7	GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition	GLoRIA	Learns global and local image-text representations by associating image sub-regions with report words; supports retrieval, classification and segmentation.	Requires paired image-report data for multimodal training and focuses on representation learning rather than a complete diagnosis-to-triage assistant.	MIMIC-CXR, CheXpert and related medical imaging datasets	https://openaccess.thecvf.com/content/ICCV2021/html/Huang_GLoRIA_A_Multimodal_Global-Local_Representation_Learning_Framework_for_Label-Efficient_Medical_Image_Recognition_ICCV_2021_paper.html
8	ConVIRT: Contrastive Learning of Medical Visual Representations from Paired Images and Text	ConVIRT	Uses bidirectional contrastive learning between paired medical images and reports, achieving strong	Depends on paired image-text data and mainly learns transferable representations rather than directly generating explanations and actionable triage.	Radiology image-report data; evaluated on multiple medical image classification and zero-shot retrieval tasks	https://arxiv.org/abs/2010.00747

			transfer performance and data efficiency.			
9	MedCLIP: Contrastive Learning from Unpaired Medical Images and Text	MedCLIP	Uses medical-knowledge-based semantic matching between images and text, improving zero-shot prediction, supervised classification and image-text retrieval.	Primarily a representation/contrastive learning framework; does not itself provide complete report generation, explanation heatmaps and edge triage.	MIMIC-CXR, CheXpert and related medical image/text resources	https://aclanthology.org/2022.emnlp-main.256/
10	ChestX-ray8: Hospital-Scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases	Weakly-supervised multi-label CNN classification/localization	Introduced a large chest X-ray benchmark and demonstrated weakly supervised detection and spatial localization of common thoracic diseases from image-level labels.	Labels are text-mined/weakly supervised; the benchmark is not a complete multimodal reasoning, report-generation or triage system.	ChestX-ray8: 108,948 frontal-view X-rays from 32,717 patients with eight text-mined disease labels	https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.html

6. Datasets and Dataset Access Links

Dataset	Purpose / Key Details	Access Link
ChestX-ray14 / NIH ChestX-ray	112,120 frontal-view chest X-rays with 14 thoracic disease labels; primary large-scale classification source.	https://nihcc.app.box.com/v/ChestXray-NIHCC
OpenI / Indiana University Chest X-ray Collection	7,470 X-rays paired with clinical radiology reports; useful for image-text alignment and report-generation experiments.	https://openi.nlm.nih.gov/
ROCO Medical Radiology Image-Text Dataset	Multimodal radiology dataset suitable for representation learning and image-caption alignment.	https://github.com/razorx89/ROCO
CheXpert	Large chest radiograph dataset with uncertainty labels and reports; useful for external evaluation/generalization.	https://stanfordmlgroup.github.io/competitions/chexpert/
MIMIC-CXR	Large chest radiograph dataset with associated free-text reports; access requires registration and credentialing/data-use requirements.	https://physionet.org/content/mimic-cxr/2.0.0/

7. Research Gap Identification

Gap 1 — Classification without integrated reasoning

Many chest-X-ray systems focus on classification. The proposed project combines disease prediction with a plain-language diagnostic description so that the output is more understandable to the intended user.

Gap 2 — Explanation disconnected from diagnosis text

Grad-CAM and saliency methods can show relevant regions, but visual explanation is often presented

separately from natural-language explanation. The proposed system links the predicted abnormality, confidence and visual evidence.

Gap 3 — Limited actionable triage

Surveyed multimodal representation models do not directly provide a practical urgency/priority layer. The project adds a conservative, rule-based triage-support module with clearly separated safety messaging.

Gap 4 — Heavy multimodal models versus rural connectivity

Vision-language models can be computationally expensive. The project explicitly targets quantization and offline Android execution rather than assuming continuous cloud access.

Gap 5 — Data limitations

Paired image-report datasets are smaller and harder to obtain than image-only datasets. The project therefore uses multiple public sources and separates pretraining/alignment data from supervised disease-classification data where appropriate.

Gap 6 — Evaluation beyond accuracy

A useful diagnostic assistant must be evaluated not only by accuracy but also by precision, recall, F1, localization/explanation quality, text factuality and edge runtime performance.

7.1 How the Project Will Address the Gaps

- Build a modular multimodal pipeline instead of a single classifier.
- Generate both visual evidence and textual explanation for each prediction.
- Keep triage logic separate from generative text so safety-critical priority decisions are deterministic and auditable.
- Use quantization and ONNX or an equivalent mobile runtime to reduce memory and inference cost.
- Use patient-wise train/validation/test splitting to reduce leakage and evaluate generalization.
- Compare the proposed system against classification-only and explanation-only baselines.

8. Innovation, Creativity and Novelty

Aspect	Project Contribution
Innovation	Integration of vision-language diagnosis, visual explanation and offline edge deployment in one chest-radiograph assistant.
Creativity	The workflow is designed around the rural health worker: upload X-ray → inspect prediction → inspect evidence → read simple explanation → see priority guidance.
Novelty	The project focuses on combining multimodal interpretation, explainability, conservative triage support and offline Android execution rather than proposing a new foundation model from scratch.
Practical Novelty	The final prototype is intended to remain useful under zero-connectivity conditions, addressing a key deployment constraint in the stated problem context.

9. Feasibility

Area	Feasibility Assessment
Technical feasibility	The project can use established open-source architectures and pretrained components. The system is modular so a smaller vision model can replace a heavy VLM stack if required.
Dataset feasibility	ChestX-ray14, OpenI and ROCO are identified as core sources, while CheXpert and MIMIC-CXR can support validation/generalization when access and scope permit.
Implementation feasibility	A staged implementation allows the team to build classification

	first, followed by explanation, language generation, triage and quantized mobile deployment.
Compute feasibility	Training/fine-tuning may require GPU resources, but pretrained encoders can be frozen and lightweight adapters/heads trained if full multimodal training is too expensive.
Deployment feasibility	INT8 quantization and ONNX/Android runtime integration are proposed. A desktop/offline prototype is a fallback if full mobile optimization is not completed.
Safety feasibility	The prototype is positioned as decision support and screening/triage assistance, not autonomous diagnosis. Outputs should expose uncertainty and recommend professional review.

10. Complete Project Workflow / Execution Plan

The project will be executed in four major phases, progressing from data preparation to model development, edge optimization, and final evaluation.

PHASE 1 — Data Preparation & Alignment

11. Collect and organize ChestX-ray14, OpenI and ROCO; add CheXpert/MIMIC-CXR only if access and scope permit.
12. Remove corrupt/duplicate files and standardize image formats.
13. Perform patient-wise train/validation/test splitting wherever patient identifiers are available.
14. Resize and normalize X-rays; apply medically appropriate augmentation to the training split only.
15. Prepare image labels for multi-label disease classification.
16. Prepare image-report pairs for vision-language alignment and report-generation experiments.
17. Create a consistent vocabulary/target format for plain-language output.

PHASE 2 — Model, Multimodal Alignment & Explainability

18. Establish a baseline CNN/DenseNet classifier.
19. Train or fine-tune the disease classification head.
20. Integrate ViT or another selected vision encoder for spatial features.
21. Add CLIP/medical contrastive alignment between image and text representations.
22. Integrate a BLIP-2-inspired generation component or a smaller image-to-text model if edge constraints require it.
23. Generate class-specific Grad-CAM heatmaps.
24. Generate cross-attention overlays when supported by the selected VLM.
25. Build the combined output: disease probabilities + explanation image + short text.

PHASE 3 — Optimization & Mobile Deployment

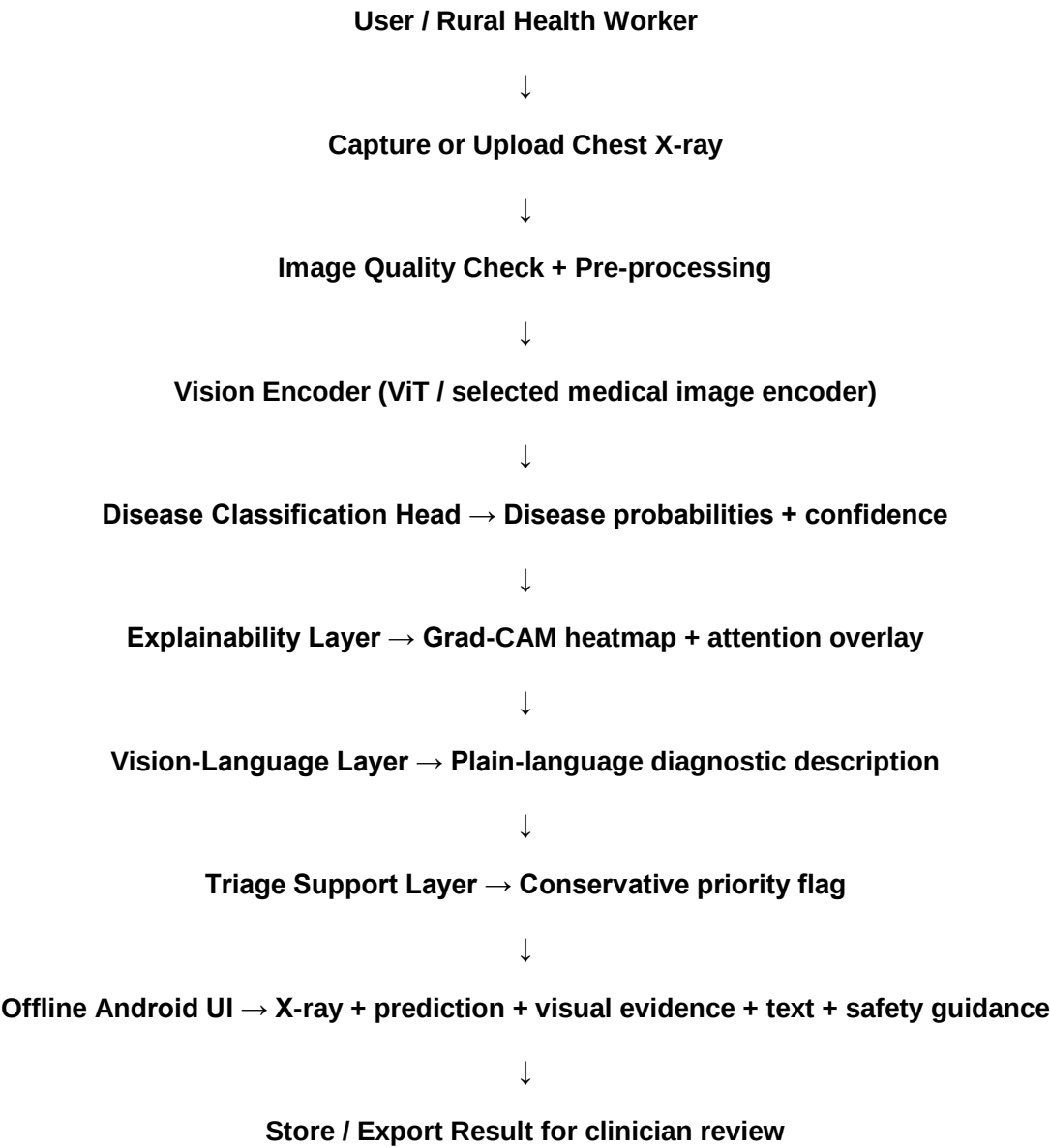
26. Profile model size, RAM use and inference latency.
27. Freeze large pretrained components where possible and reduce trainable parameters.
28. Apply FP16/INT8 quantization and verify that predictive performance remains acceptable.
29. Export compatible components to ONNX or an Android-supported inference runtime.
30. Build an Android interface for X-ray input and results.
31. Implement local storage and offline inference without requiring an API call.
32. Add safety messages and clearly separate model prediction from triage-support rules.

PHASE 4 — Evaluation, Validation & Presentation

33. Evaluate classification using Precision, Recall, F1 and ROC-AUC, with accuracy where appropriate.
34. Evaluate explanation quality using localization checks where ground-truth regions are available and/or expert qualitative review.
35. Evaluate generated text using task-appropriate language metrics and factual consistency review.

- 36. Measure mobile latency, memory footprint and model size.
- 37. Run ablation studies: classifier only vs classifier + Grad-CAM; image-only vs image-text alignment; full precision vs INT8.
- 38. Test representative normal and abnormal cases and document failure cases.
- 39. Prepare final demo, report, architecture diagram, evaluation tables and presentation.

10.1 End-to-End System Flow



11. Evaluation Strategy

Component	Metrics	Evaluation Goal
Disease classification	Precision, Recall, F1-score, ROC-AUC, sensitivity/specificity where applicable	Compare with DenseNet/CheXNet-style baseline.
Visual explanation	Localization overlap/pointing-game style checks where annotations exist; qualitative expert review	Check whether highlighted regions correspond to plausible pathology regions.
Text generation	Clinical factuality review plus standard text-similarity metrics where suitable	Penalize unsupported findings and hallucinated diagnoses.
Triage	Sensitivity for high-priority cases; false-negative analysis; rule consistency	Triage should be conservative and never be presented as definitive diagnosis.

Edge deployment	Latency, RAM, model size, battery/runtime behavior	Compare FP32, FP16 and INT8 versions.
Robustness	Performance across datasets/subgroups where metadata permits	Investigate distribution shift and failure cases.

12. Project Plan / Timeline

Period	Work Package
Week 1	Dataset acquisition, cleaning, patient-wise split, preprocessing pipeline
Week 2	Baseline DenseNet/ResNet classifier and evaluation
Week 3	ViT/CLIP-style multimodal alignment experiments
Week 4	BLIP-2-inspired report generation / lightweight alternative
Week 5	Grad-CAM and attention visualization integration
Week 6	Triage-support logic, safety layer and integrated backend
Week 7	INT8 quantization, ONNX/mobile runtime optimization
Week 8	Android UI, offline testing and performance profiling
Week 9	Ablation studies, robustness testing and error analysis
Week 10	Final evaluation, documentation, demo and presentation

13. Expected Final Deliverables

- Working offline/edge prototype for chest X-ray decision support.
- Multi-label disease predictions with confidence scores.
- Visual Grad-CAM/attention explanations.
- Plain-language diagnostic descriptions.
- Conservative triage/priority indicator with safety disclaimer.
- Android/mobile demonstration or equivalent offline prototype if device constraints require it.
- Experimental comparison against baseline models.
- Complete project report containing methodology, literature survey, research gap, experiments, results and limitations.

14. Anticipated Limitations and Risk Mitigation

Risk / Limitation	Mitigation
Dataset label noise	ChestX-ray14 labels are largely derived from reports; use patient-wise splits, external validation and error analysis.
Model hallucination	Constrain generated text to detected findings/templates where possible and add factual consistency checks.
Explanation faithfulness	Treat heatmaps as model-attribution evidence, not proof of pathology; validate localization and present uncertainty.
Mobile compute limits	Use frozen encoders, smaller models, quantization and staged deployment.
Distribution shift	Evaluate on an external dataset when possible and report limitations instead of claiming universal performance.
Clinical safety	Do not position the prototype as an autonomous diagnostic device; require clinician confirmation for medical decisions.

15. Key References and Dataset Links

- BLIP-2 — <https://arxiv.org/abs/2301.12597>
- MedKLIP — <https://arxiv.org/abs/2301.02228>
- Grad-CAM — <https://arxiv.org/abs/1610.02391>

- CheXNet — <https://arxiv.org/abs/1711.05225>
- GLoRIA — https://openaccess.thecvf.com/content/ICCV2021/html/Huang_GLoRIA_A_Multimodal_Global-Local_Representation_Learning_Framework_for_Label-Efficient_Medical_Image_Recognition_ICCV_2021_paper.html
- ConVIRT — <https://arxiv.org/abs/2010.00747>
- MedCLIP — <https://aclanthology.org/2022.emnlp-main.256/>
- ChestX-ray8 — https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.html
- DenseNet — <https://arxiv.org/abs/1608.06993>
- ResNet — <https://arxiv.org/abs/1512.03385>
- NIH ChestX-ray14 — <https://nihcc.app.box.com/v/ChestXray-NIHCC>
- OpenI — <https://openi.nlm.nih.gov/>
- ROCO — <https://github.com/razorx89/ROCO>
- CheXpert — <https://stanfordmlgroup.github.io/competitions/chexpert/>
- MIMIC-CXR — <https://physionet.org/content/mimic-cxr/2.0.0/>